MEA CO2 Capture Ver 1 · operator training simulator

DOCUMENT REGISTER

Register of the documents issued by the OTS, deliverable tree, project phases, engineering queries

MEA CO2 capture unit: absorber T-101 / stripper T-102 (30 wt% MEA)

Document no.	MEA-PM-DOC-001
Revision	0 · Issued for teaching
Date	28-Oct-2026
Drawings	CC-PID-101 rev A · MEA-PFD-001 rev 0
Originator	Apinan Soottitantawat
Checker / approver	— (not checked or approved)

Revision record

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
0	28-Oct-2026	Issued for teaching	Apinan Soottitantawat		

Teaching document generated for the MEA CO2 capture operator training simulator. Not for construction. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

Contents

1 · Purpose, numbering and status . . . 2
2 · Document register . . . 3
3 · Deliverable tree by discipline . . . 4
4 · Project phases and where each document is issued . . . 6
5 · Engineering queries register . . . 7
6 · Queries by document and discipline . . . 12
7 · Checks . . . 13

Body pages are A4 landscape. The register, the tree, the phases and the queries are read from one data file (reg.json) that also feeds the Project & deliverables tab of the OTS, which recomputes the checks of part 7 in the page.

1 · Purpose, numbering and status

Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

This register lists every document the MEA CO2 Capture Ver 1 operator training simulator issues, the tab of the OTS that teaches it and the file published beside the page (files/...). Beside it, part 5 holds the engineering queries: each inconsistency one tab found between two documents, with the documents in conflict, a proposed resolution and a status. A query closes when a revision of the documents removes the conflict.

Numbering: **MEA-<discipline>-<type>-001**; the PFD is MEA-PFD-001 and the P&ID keeps its sheet number CC-PID-101. Revision codes follow typical practice: letters (A, B ...) before approval, numbers (0, 1 ...) from the first approved issue. Every document here is "Issued for teaching": originator Apinan Soottitantawat, checker and approver blank.

Code	Discipline	Documents in this register
PM	Project management	MEA-PM-DOC-001
PR	Process	MEA-PFD-001, MEA-PE-HMB-001, CC-PID-101, MEA-PE-RLB-001
SA	Safety	MEA-SA-HZP-001
IN	Instrumentation	MEA-IN-SRS-001, MEA-IN-CE-001, MEA-IN-INL-001
ME	Mechanical	MEA-ME-EQL-001
EE	Electrical	MEA-EE-LL-001
CM	Commissioning	MEA-CM-PLN-001

2 · Document register

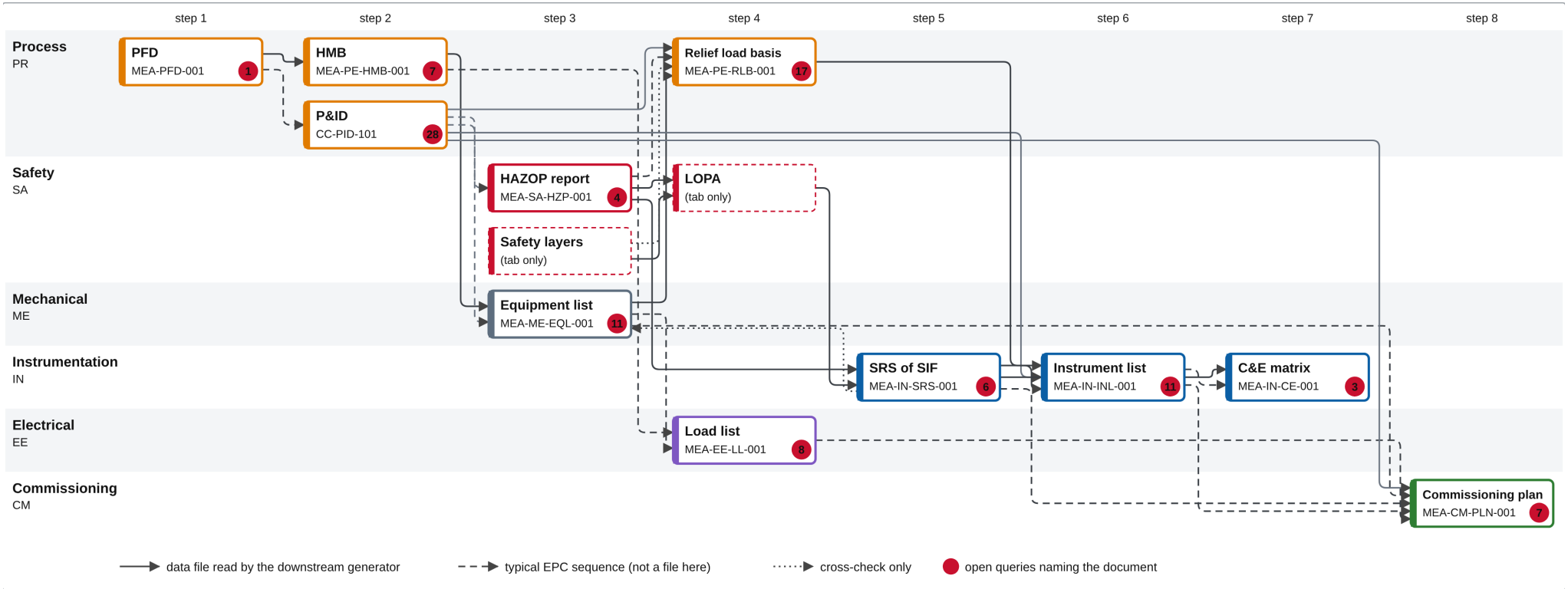
#	Document no.	Title	Disc.	Rev	Status	Tab in the OTS	Files (at files/)	Pages	Open EQ
001	**MEA-PM-DOC-001**	Document register and engineering queries	PM	0	Issued for teaching	Project & deliverables	MEA-PM-DOC-001.pdf	13	0
002	**MEA-PFD-001**	Process flow diagram	PR	0	Issued for teaching	PFD	MEA-PFD-001.json, MEA-PFD-001.pdf, MEA-PFD-001.png, MEA-PFD-001.pptx, MEA-PFD-001.svg	1	1
003	**MEA-PE-HMB-001**	Heat and material balance, line sizing and pump hydraulics	PR	0	Issued for teaching	Process engineering	MEA-PE-HMB-001.pdf	16	7
004	**CC-PID-101**	Piping and instrumentation diagram, absorber T-101 / stripper T-102	PR	A	Issued for OTS (training)	P&ID	CC-PID-101.pdf, CC-PID-101.png, CC-PID-101.svg	1	28
005	**MEA-SA-HZP-001**	HAZOP study report	SA	A	Issued for teaching	HAZOP	MEA-SA-HZP-001.pdf	67	4
006	**MEA-IN-SRS-001**	Safety requirements specification of the safety instrumented functions	IN	0	Issued for teaching (in tab, no PDF)	SRS of SIF	— (no file)	—	6
007	**MEA-IN-CE-001**	Cause and effect matrix · SIS interlocks I-1 / I-2 / I-3 and DCS alarms	IN	0	Issued for teaching	C&E matrix	MEA-IN-CE-001.pdf	2	3
008	**MEA-ME-EQL-001**	Equipment list and mechanical datasheets	ME	0	Issued for teaching	Mechanical	MEA-ME-EQL-001.pdf	14	11
009	**MEA-PE-RLB-001**	Relief load basis · PSV, TSV and vent register of the capture unit	PR	0	Issued for teaching	Relief & vent	MEA-PE-RLB-001.pdf	12	17
010	**MEA-IN-INL-001**	Instrument list · I/O count · control-valve check	IN	0	Issued for teaching	Instrumentation & control	MEA-IN-INL-001.pdf	8	11
011	**MEA-EE-LL-001**	Electrical load list	EE	0	Issued for teaching	Electrical	MEA-EE-LL-001.pdf	5	8
012	**MEA-CM-PLN-001**	Procurement to commissioning plan	CM	0	Issued for teaching	Procurement to commissioning	MEA-CM-PLN-001.pdf	7	7

Studies held only in a tab, with no document number in this register: LOPA worksheet (6 scenarios, SIF credits) (LOPA tab); Safety layers register (Safety layers tab). Queries also cite the OTS model and HMI ("OTS model / HMI"); they are sources, not documents.

Build list: 17 files published at files/ beside the page: MEA-IN-CE-001.pdf (2 p), MEA-CM-PLN-001.pdf (7 p), MEA-EE-LL-001.pdf (5 p), MEA-IN-INL-001.pdf (8 p), MEA-ME-EQL-001.pdf (14 p), MEA-PE-HMB-001.pdf (16 p), MEA-PM-DOC-001.pdf (13 p), MEA-PE-RLB-001.pdf (12 p), MEA-SA-HZP-001.pdf (67 p), MEA-PFD-001.json, MEA-PFD-001.pdf (1 p), MEA-PFD-001.png, MEA-PFD-001.pptx, MEA-PFD-001.svg, CC-PID-101.pdf (1 p), CC-PID-101.png, CC-PID-101.svg.

3 · Deliverable tree by discipline

Which document feeds which. Solid arrow: the generator of the downstream document reads the upstream document's data file (found by scanning each tab's tools). Dashed: the typical EPC sequence, not wired as a file in this OTS. Dotted: read only for a cross-check. The red badge counts the open queries naming the document.



Links of the tree

From	To	Kind	From	To	Kind
Equipment list	Load list	typical sequence	PFD	P&ID	typical sequence
Equipment list	Commissioning plan	typical sequence	P&ID	Equipment list	typical sequence
Equipment list	Relief load basis	data file	P&ID	HAZOP report	typical sequence
HMB	Equipment list	data file	P&ID	Instrument list	data file
HMB	Load list	typical sequence	P&ID	Commissioning plan	data file
HAZOP report	LOPA	data file	P&ID	Relief load basis	data file

From	To	Kind	From	To	Kind
HAZOP report	Relief load basis	typical sequence	Relief load basis	Instrument list	data file
HAZOP report	SRS of SIF	data file	Safety layers	LOPA	data file
Instrument list	C&E matrix	typical sequence	Safety layers	Relief load basis	cross-check
Instrument list	Commissioning plan	typical sequence	SRS of SIF	C&E matrix	data file
Load list	Commissioning plan	typical sequence	SRS of SIF	Equipment list	cross-check
LOPA	SRS of SIF	data file	SRS of SIF	Instrument list	data file
PFD	HMB	data file	SRS of SIF	Commissioning plan	typical sequence

Cross-check links: SRS of SIF → Equipment list: the EQL copies the SRS blocked-outlet run for its check / EQ-M03; Safety layers → Relief load basis: the RLB checks its devices against the Safety layers register.

4 · Project phases and where each document is issued

I = first issue, R = revised (for example issued for construction, as-built), U = used as an input in that phase. Typical EPC practice, not a schedule. The OTS documents sit at rev 0 "Issued for teaching", which corresponds to the end of FEED / early detailed engineering: the HOLDs and open queries of part 5 are what a detailed-engineering revision would close.

Document	Title	FEED	Detailed engineering	Procurement	Construction	Commissioning
MEA-PM-DOC-001	Document register and engineering queries	**I**	**R**	**R**	**R**	**R**
MEA-PFD-001	Process flow diagram	**I**	**R**			**U**
MEA-PE-HMB-001	Heat and material balance, line sizing and pump hydraulics	**I**	**R**	**U**		**U**
CC-PID-101	Piping and instrumentation diagram, absorber T-101 / stripper T-102	**I**	**R**		**U**	**R**
MEA-SA-HZP-001	HAZOP study report	**I**	**R**			**U**
MEA-IN-SRS-001	Safety requirements specification of the safety instrumented functions		**I**	**U**		**U**
MEA-IN-CE-001	Cause and effect matrix · SIS interlocks I-1 / I-2 / I-3 and DCS alarms		**I**	**U**		**U**
MEA-ME-EQL-001	Equipment list and mechanical datasheets	**I**	**R**	**U**		
MEA-PE-RLB-001	Relief load basis · PSV, TSV and vent register of the capture unit		**I**	**U**		**U**
MEA-IN-INL-001	Instrument list · I/O count · control-valve check		**I**	**U**	**U**	**U**
MEA-EE-LL-001	Electrical load list		**I**	**U**	**U**	
MEA-CM-PLN-001	Procurement to commissioning plan			**I**	**R**	**U**
First issues		6	5	1	0	0

5 · Engineering queries register

Collected from the "Engineering queries" entries of the tabs' NOTES (Process engineering, Mechanical, Relief & vent, Instrumentation & control, Electrical, Procurement to commissioning, the P&ID engineering view) and the safety findings of the HAZOP, LOPA, SRS and Safety layers tabs (EQ-SA-xx). A resolution in *italics* is proposed by this register because the source gives none. Status Agreed = the documents now agree; the query closes at the next revision.

EQ	Raised by	Documents in conflict	Issue	Proposed resolution	Status
EQ-P01	Process engineering Process	CC-PID-101 MEA-PE-HMB-001	**Line sizes on CC-PID-101 (all marked TBC) disagree with the sizing in part 4** CC-PID-101 rev A vs MEA-PE-HMB-001 Amine lines drawn DN400 need DN250–300; VR-302 DN500 gives 42 m/s (needs DN700); LS-207 DN400 → DN500; WW-105 DN200 → DN300; CO-304 DN300 → DN400.	Issue the line list with the sizes of part 4 and update the P&ID at rev B.	**Open**
EQ-P02	Process engineering Process	MEA-PFD-001 MEA-PE-HMB-001 also: OTS model / HMI	**Liquid pressures in the HMB/PFD (model allowances) are below the pump hydraulics** MEA-PFD-001 flags, Process tab vs part 5 Stream 11 (E-101 rich outlet) 295 vs 400 kPa a; stream 17 (reflux) 325 vs 463 kPa a; stream 7 (FV-201 inlet) 403 vs 453 kPa a.	Keep the model allowances for the OTS; mark them "allowance" on the PFD and take pump-side pressures from MEA-PE-HMB-001.	**Open**
EQ-P03	Process engineering Process	CC-PID-101 MEA-PE-HMB-001 also: OTS model / HMI	**Hot rich amine flashes after E-101** Model VLE vs CC-PID-101 (FV-109 upstream of E-101) Model bubble pressure of the rich solvent at 109 °C, α 0.471 is 1352 kPa a, above the line pressure (400 kPa a): the rich side of E-101 and RA-104 run two-phase. The correlation is extrapolated (TBC) but flashing here is normal practice.	Move the rich letdown (FV-109) to the T-102 inlet or add a back-pressure valve there; size RA-104 for two-phase flow; confirm with measured VLE.	**Open**
EQ-P04	Process engineering Process	MEA-PE-HMB-001 also: OTS model / HMI	**The 6 % CO2 case sets the pumps and P-102 has no margin on FV-201** Part 6 vs model valve capacity Lean flow at 6 % CO2 is 473 t/h (1.76 × design); rated P-102 521 t/h equals FV-201 wide open (K.FmaxL 520 t/h). At design P-102 runs at 52 % of rated flow.	Confirm the 6 % CO2 design basis; consider 2 × 60 % pumps and a larger FV-201 trim; check the minimum continuous flow.	**Open**
EQ-P05	Process engineering Process	MEA-PE-HMB-001 also: OTS model / HMI	**Cooling water saturates at 6 % CO2** Part 6 (case B) TV-202 and TV-302 reach 100 % and 100 %; lean to T-101 42.4 °C (TIC-202 SP 40). The CW flows in the model are assumed (1000/400/500 kg/s) and give a CW rise of only 1.5 K in E-104 at design.	Set CW flows for a typical 8–10 K rise and size E-103/E-104 for case B; update the model constants at the next model revision.	**Open**
EQ-P06	Process engineering Process	MEA-PE-HMB-001	**P-102 NPSH needs an elevated kettle** Part 5 vs layout NPSHa = static head only (saturated lean). E-102 low level must be at ≥ 5.66 m above grade (5.90 m used) and the T-102 bottom tangent above the kettle liquid level.	Fix E-102 at +5.5 m and T-102 at +9.5 m in the Layout tab and the GA (Mechanical tab).	**Open**
EQ-M01	Mechanical Mechanical	MEA-ME-EQL-001 MEA-PE-RLB-001	**E-102 tube rupture is a relief case** MEA-ME-EQL-001 vs relief basis (HOLD) Steam side 420 kPa g vs process side 270 kPa g: 1.3 × 270 = 351 < 420, so the 10/13 rule (API 521) does not exclude a tube rupture; steam would flow into the T-102 system.	Add tube rupture to PSV-301 sizing (Relief & vent tab) or raise the T-102 system design pressure to ≥ 330 kPa g.	**Open**
EQ-M02	Mechanical Mechanical	MEA-ME-EQL-001	**E-103 and E-104 tube rupture** EQL vs CW system (assumed 400 kPa a) E-103: CW 470 kPa g vs process 270 kPa g (CW into the T-102 system). E-104: lean side 770 kPa g vs CW 470 kPa g (amine into the CW return).	Relief cases for PSV-301 and for the CW return of E-104, or raise the low-side design pressures; confirm the CW supply pressure.	**Open**
EQ-M03	Mechanical Mechanical	CC-PID-101 MEA-ME-EQL-001 MEA-IN-SRS-001 also: OTS model / HMI	**PSV-301 set vs blocked-outlet pressure** CC-PID-101 note 5 (HOLD), SRS MEA-IN-SRS-001, EQL With every trip disabled the model reaches ≥ 400 kPa a on a blocked CO2 outlet (189 s to 350 kPa a); the T-102 system MAWP 270 kPa g allows 398 kPa a at 10 % accumulation. PSHH-301 / I-2 alone does not stop the rise (SRS SIF-2 gap).	Size PSV-301 for blocked outlet with the hot rich feed flashing (Relief & vent tab); keep set ≤ MAWP; add SIF-2R (stop P-101) as the LOPA recommends.	**Open**

EQ	Raised by	Documents in conflict	Issue	Proposed resolution	Status
EQ-M04	Mechanical Mechanical	CC-PID-101 MEA-ME-EQL-001 also: Theory tab	**Materials: P&ID note 6 vs the equipment list** CC-PID-101 note 6 vs MEA-ME-EQL-001, Theory tab Note 6 allows "CS + CA for cold lean service"; the EQL uses SS 316L for every amine-wetted part (E-104 shell, P-102 casing, T-101 shell) as the Theory tab states.	Keep SS 316L; revise note 6 at rev B, or justify CS + CA by a corrosion study (loading, velocity, heat-stable salts).	**Open**
EQ-M05	Mechanical Mechanical	CC-PID-101 MEA-ME-EQL-001 also: Safety layers tab	**Which side of E-102 the HOLD PSV protects** CC-PID-101 note 5 vs EQL, Safety layers register Note 5 lists a PSV on the "E-102 shell". In a kettle the shell is the process side, open to T-102 and covered by PSV-301; the steam tubes need their own PSV-207.	Confirm PSV-207 on the steam side; delete the shell PSV or record it as PSV-301.	**Open**
EQ-M06	Mechanical Mechanical	MEA-ME-EQL-001	**Mechanical checks not done** EQL Full vacuum (external pressure, stiffening rings), wind and seismic on T-101 (35 m) and T-102 (24 m), nozzle loads and the T-101 skirt are not checked; T-101 thickness is set by the rigidity minimum.	HOLD for the vessel vendor; check UG-28 for the FV items before issue.	**Open**
EQ-M07	Mechanical Mechanical	CC-PID-101 MEA-ME-EQL-001	**Equipment not on the P&ID** CC-PID-101 note 7 vs EQL No solvent filters (mechanical, carbon), reclaimer, amine storage or drain sump are drawn, so the EQL has none; an MEA unit needs them (foaming and degradation are faults in the simulator).	Add them at P&ID rev B and to the EQL; the filter ΔP then enters the P-101 hydraulics.	**Open**
EQ-M08	Mechanical Mechanical	MEA-ME-EQL-001 also: OTS model / HMI	**Model UA vs datasheet U·A** src/model.js K vs EQL E-101 model UA 2000 kW/K = 63 % of the datasheet U·A (sized for the 6 % CO2 case); E-103 68 %, E-104 51 %. The model's exchangers are smaller than the datasheets.	Accept for training (the model reproduces the design point); note that the datasheet over-surface is spent at 6 % CO2.	**Open**
EQ-R01	Relief & vent Process	CC-PID-101 MEA-PE-RLB-001 also: Safety layers tab	**P&ID note 5: "PSV … E-102 SHELL"** CC-PID-101 note 5; Safety layers (PSV-207); this RLB In a kettle reboiler the shell is the boiling process side, which is open to T-102, so PSV-301 covers it. The steam is in the tube bundle; whether that side needs a PSV is a different question (PSV-207, EQ-R06). related: EQ-R06	Revise note 5: "PSV-301 (T-102 overhead) protects T-102, E-102 shell (process side), E-103 and V-101; E-102 tube side see PSV-207 decision; TSV-501…504 thermal".	**Open**
EQ-R02	Relief & vent Process	MEA-ME-EQL-001 MEA-PE-RLB-001 also: OTS model / HMI	**PSV-301 set vs MAWP (now agree)** HMI text of PSHH-301 (PSV-301 at 350 kPa a); EQL MEA-ME-EQL-001 (T-102 system 270 kPa g = 371.3 kPa a); this RLB The EQL now issues the T-102 system design pressure. The set stays at 350 kPa a, below the MAWP, as every other tab uses. Relieving at set + 10 % and fire at set + 21 % stay inside 110 % / 121 % of the MAWP (398.3 / 428.0 kPa a). related: EQ-M03	Agreed. Keep the set at 350 kPa a and show PSV-301 on CC-PID-101 with the EQL design pressure. This closes the open part of EQL EQ-M03 (blocked outlet within 110 % of the MAWP once PSV-301 is sized for R1).	**Agreed**
EQ-R03	Relief & vent Process	MEA-PE-RLB-001 also: OTS model / HMI	**Model pressure clamp** src/model.js (P.Pd clamped to 95–400 kPa a); this RLB (fire relieving 402 kPa a) The model cannot represent the drum above 400 kPa a, so the fire case (21 % overpressure) is computed by the API 521 equation, not by a model run.	Keep the clamp: other tabs store runs that depend on it. The fire case stays an API 521 hand calculation in this RLB.	**Open**
EQ-R04	Relief & vent Process	MEA-PE-RLB-001 MEA-IN-SRS-001 also: LOPA tab	**I-2 steam trip does not end a blocked outlet** SRS (SIF-2 gap); LOPA (SIF-2R); this RLB run R3 With PV-301 closed and XV-207 already shut, the hot rich feed still flashes and lifts PSV-301 (run R3). The relief device is the last layer for this case.	Keep PSV-301 sized for R1. Close the LOPA action SIF-2R (stop P-101 on PSHH-301), which removes the flashing feed.	**Open**
EQ-R05	Relief & vent Process	CC-PID-101 MEA-PE-RLB-001	**Isolation of V-101 from T-102** CC-PID-101 (no block valve drawn); this RLB (PSV-302 not required) PSV-302 is dropped only because nothing can isolate V-101.	Confirm on the P&ID and the isometrics that no block valve is fitted; otherwise add PSV-302 (fire + blocked outlet).	**Open**

EQ	Raised by	Documents in conflict	Issue	Proposed resolution	Status
EQ-R06	Relief & vent Process	MEA-ME-EQL-001 MEA-PE-RLB-001 MEA-IN-SRS-001 also: Safety layers tab	**E-102 steam-side rating vs letdown failure (disagree)** EQL MEA-ME-EQL-001 (E-102 tubes 420 kPa g, from the LP header pressure); SRS SIF-3 demand case (10 bar a MP steam); Safety layers (PSV-207 Gap); this RLB With the EQL rating a letdown failure overpressures the steam side, and PSV-207 sized for FV-207 wide open on MP steam needs several T orifices (part 4).	Re-rate the E-102 tube side and the FV-207 downstream piping for the MP header in the EQL (then no PSV-207), or confirm a full-flow relief on the LP header at the letdown station that limits it to the EQL rating.	**Open**
EQ-R07	Relief & vent Process	CC-PID-101 MEA-PE-RLB-001	**P&ID note 5 lists a PSV on E-101** CC-PID-101 note 5; this RLB E-101 is a plate exchanger with liquid on both sides: a plate leak is not an overpressure case. Only thermal expansion of the blocked-in rich side needs a relief device.	Show TSV-501 on the rich side; delete "E-101" from the PSV list of note 5.	**Open**
EQ-R08	Relief & vent Process	MEA-IN-INL-001 MEA-PE-RLB-001 also: OTS model / HMI	**Steam valve pressure drop in the model** src/model.js (reboiler condenses at the header pressure); MEA-IN-INL-001 (FV-207 sizing) Because FV-207 takes no pressure drop in the model, the reboiler pinch at relieving pressure is the most favourable it can be. A real valve ΔP lowers the condensing temperature and the blocked-outlet load.	Keep the model load (conservative). Recompute when the steam-side hydraulics are issued.	**Open**
EQ-R09	Relief & vent Process	MEA-PE-RLB-001 also: Layout tab (not built)	**Fire wetted area uses assumed elevations** This RLB (fire case); Layout & isometrics (not issued) The fire load depends on how much of T-102 and V-101 lies within 7.6 m of grade.	Confirm the elevations in the Layout tab and recompute (the fire calculator in part 3 takes them as inputs).	**Open**
EQ-R10	Relief & vent Process	CC-PID-101 MEA-PE-RLB-001 also: Safety layers tab, Consequence tab	**No CO2 vent on the P&ID** CC-PID-101; Safety layers (VENT-CO2); Consequence tab (CO2 dense plume) The product line has no start-up / compressor-trip vent and the PSV discharge location is not shown.	Add HV-305 and the vent stack VS-301 (shared with PSV-301) at a high point; check dispersion in the Consequence tab.	**Open**
EQ-R11	Relief & vent Process	MEA-ME-EQL-001 MEA-PE-RLB-001	**E-102 and E-103 tube rupture (agree: credible, not governing)** EQL MEA-ME-EQL-001 EQ-M01, EQ-M02; this RLB (part 3) Both are credible by the 10/13 rule with the EQL design pressures. Steam at the normal LP pressure cannot enter a shell relieving at set + 10 % (zero load). Cooling water does enter E-103, but as liquid it only displaces vapour, so its relief load is small. Neither governs PSV-301. related: EQ-M01, EQ-M02	Agreed with the EQL that they are relief cases: rows TR and TR2 of the PSV-301 case table. Confirm the CW supply pressure (the EQL assumes 400 kPa a).	**Agreed**
EQ-R12	Relief & vent Process	MEA-ME-EQL-001 MEA-PE-RLB-001	**E-104 tube rupture: amine into the CW side (partly disagree)** EQL MEA-ME-EQL-001 EQ-M02 (relief case for the CW return); this RLB Credible by the 10/13 rule, but at its maximum operating pressure the lean side cannot exceed the CW-side MAWP, so no relief load results. Overpressure would need P-102 at shut-off with FV-201 closed as well (two failures). related: EQ-M02	No relief device for this case. Add amine detection on the CW return (TBC) and keep TSV-502 for thermal expansion only.	**Open**
EQ-I01	Instrumentation & control Instrumentation	MEA-IN-INL-001 also: OTS model / HMI	**FIC-201 span** makeCtrls() FIC-201 0–500 t/h; DCS trend range CRANGE 0–400 t/h The controller works on 0–500 t/h but the trend and faceplate scale is 0–400 t/h. A transmitter calibrated to one and a display scaled to the other read the same mA as two different flows.	Calibrate FT-201 0–500 t/h (it must cover the FV-201 capacity) and set the DCS range to the same.	**Open**
EQ-I02	Instrumentation & control Instrumentation	CC-PID-101 MEA-IN-INL-001 also: OTS model / HMI	**Analyzer tag letters** Screen bubbles AI-101/111/113/203; CC-PID-101 AIT-101/111/113/203; AIT-213 on both One instrument carries two tags in two documents.	Use AIT-xxx for the transmitter and AI-xxx (or AIC) for the DCS indication; update the screen or the P&ID.	**Open**
EQ-I03	Instrumentation & control Instrumentation	CC-PID-101 MEA-IN-INL-001 also: OTS model / HMI	**Tags not on the P&ID** Screen / IND: FI-208, FI-303, TI-401; AY-212, AY-214 only in note 8 Instruments that the DCS shows have no bubble on CC-PID-101.	Add FT-208, FT-303 and TT-401 to CC-PID-101 at the next revision.	**Open**

EQ	Raised by	Documents in conflict	Issue	Proposed resolution	Status
EQ-I04	Instrumentation & control Instrumentation	MEA-IN-INL-001 also: OTS model / HMI	**FV-208 oversized; FI-208 range** Model K.FmaxMU 5 t/h, FV-208 at about 5 % open at design; IND FI-208 0–1 t/h The make-up valve works at the bottom of its stroke, where control is poor, and the flowmeter cannot read the flow when the valve is wide open.	Size FV-208 for about 1 t/h (design opening 50–70 %) or add a small trim; re-range FI-208 to cover the valve capacity.	**Open**
EQ-I05	Instrumentation & control Instrumentation	MEA-IN-INL-001 also: OTS model / HMI	**Cooling-water flows in the model** K.Wcwl / Wcww / Wcwc (CW valve capacities) vs exchanger duties At design the CW temperature rise is only 1.5–5.5 K (E-104 1.5 K), so the CW valve Cv in this list is much larger than a real CW balance (8–10 K rise, typical) would give.	Size TV-202 / TV-110 / TV-302 from the CW balance of the utility study; keep the model values for training only.	**Open**
EQ-I06	Instrumentation & control Instrumentation	MEA-PE-RLB-001 MEA-IN-INL-001 also: OTS model / HMI	**Pressure ranges vs the PSV set** IND PSHH-301 and PIC-301 0/100–300 kPa a; MEA-PE-RLB-001 PSV-301 set 350, relieving 375 kPa a Neither PT-301 nor PT-301B can show the approach to the PSV set or the relieving pressure; they peg at 300 kPa a.	Range PT-301 and PT-301B 0–400 kPa a.	**Open**
EQ-I07	Instrumentation & control Instrumentation	MEA-PE-RLB-001 MEA-IN-INL-001 also: OTS model / HMI	**Steam valve ΔP** Model (reboiler condenses at the header pressure); MEA-PE-RLB-001 EQ-R08 FV-207 is sized here with an assumed 50 kPa drop, which the model does not have. related: EQ-R08	Confirm the LP header pressure at E-102 and recompute FV-207.	**Open**
EQ-I08	Instrumentation & control Instrumentation	MEA-IN-INL-001 also: OTS model / HMI	**Flow ranges below valve capacity** IND FI-109 0–500 vs FV-109 545 t/h; FIC-207 0–40 vs FV-207 47 t/h A wide-open valve drives the transmitter over range, so the operator loses the reading in the upset where it matters.	Range each flow transmitter to at least the valve capacity, or record the reason (e.g. pump curve limit).	**Open**
EQ-EE-01	Electrical Electrical	CC-PID-101 MEA-EE-LL-001 also: OTS model / HMI	**K-101 motor 355 kW vs 589 kW at full speed** Load list MEA-EE-LL-001; P&ID CC-PID-101 equipment list (313 kW); model FIC-101 range 0–250 kNm³/h, blower curve K-101 motor 355 kW (design shaft 313 kW × 1.10) is below the model's blower power at 100 % VFD speed (589 kW at 223 kNm³/h).	Either cap the VFD speed (and FIC-101 output) so the motor cannot be overloaded, or size the motor for the full curve (next standard 630 kW).	**Open**
EQ-EE-02	Electrical Electrical	CC-PID-101 MEA-EE-LL-001	**No spare pumps on the P&ID** CC-PID-101 equipment list vs load list No spare pumps on the P&ID; the load list carries P-101B…P-104B as TBC standby (no demand).	Confirm the sparing philosophy (2 × 100 % is typical for amine circulation).	**Open**
EQ-EE-03	Electrical Electrical	MEA-PE-HMB-001 MEA-EE-LL-001 also: OTS model / HMI	**Pump ΔP in no issued document** Load list vs HMB / Process engineering Pump differential pressures are not in any issued document; the model has no pump curves. The load list builds them from model pressures plus assumed elevations and equipment ΔP.	Replace with the pump hydraulic calculation from the Process engineering tab when issued.	**Open**
EQ-EE-04	Electrical Electrical	MEA-SA-HZP-001 MEA-EE-LL-001	**Hot MEA above the flash point of pure MEA** Hazardous-area note in this load list; HAZOP; SDS Lean MEA is handled at 120 °C (TIC-206), above the 96 °C flash point of pure MEA; the flash point of the 30 wt% aqueous solution is not known.	Test the solution flash point; finish the IEC 60079-10-1 assessment before the motor and instrument Ex ratings are frozen.	**Open**
EQ-EE-05	Electrical Electrical	CC-PID-101 MEA-EE-LL-001 also: OTS model / HMI	**Wash-water circulation not modelled** Model vs CC-PID-101 (E-105 5.8 MW) vs load list Wash-water circulation (P-104) is not modelled; 300 m³/h is assumed from the E-105 duty on the sheet.	Set the wash flow in the HMB.	**Open**
EQ-EE-06	Electrical Electrical	CC-PID-101 MEA-EE-LL-001 also: OTS model / HMI	**CW pumps: battery limit not defined** Model (sw.CWP) vs CC-PID-101 vs load list The model has a cooling-water supply switch (CWP) but CW pumps are on neither the P&ID nor this load list.	Define the battery limit: CW pumps by the host utility (assumed) or in this unit.	**Open**

EQ	Raised by	Documents in conflict	Issue	Proposed resolution	Status
EQ-CM-01	Procurement to commissioning Commissioning	CC-PID-101 MEA-PE-RLB-001 MEA-CM-PLN-001	**PSVs on HOLD before leak test / RFSU** CC-PID-101 note 5; this plan PC-04; Relief & vent PSVs on T-102, V-101, E-102 shell and E-101 are on HOLD; a system cannot be leak-tested above its relief set or reach RFSU without its relief devices.	Close the relief load basis and install the PSVs before PC-04 / RFSU of SYS-04 and SYS-06 (punch A).	**Open**
EQ-CM-02	Procurement to commissioning Commissioning	CC-PID-101 MEA-CM-PLN-001	**No N2 connections drawn** CC-PID-101 (no N2) vs this plan No nitrogen connection is drawn, yet purging and blanketing (PC-09) need one on T-102 / V-101 and the amine storage.	Add N2 utility connections at rev B.	**Open**
EQ-CM-03	Procurement to commissioning Commissioning	CC-PID-101 MEA-CM-PLN-001	**Amine storage and make-up not shown** CC-PID-101 note 7 vs PC-10 Amine storage, make-up pump and filter are not shown (note 7) but the first fill (PC-10) depends on them.	Issue the solvent-management sheet before procurement of PKG-09.	**Open**
EQ-CM-04	Procurement to commissioning Commissioning	MEA-CM-PLN-001	**Performance-test criteria in no issued document** This plan part 7 vs the design basis Performance-test acceptance criteria (capture, SRD, product, duration) are not in any issued document; this plan assumes them.	Agree the guarantee values and test procedure in the contract.	**Open**
EQ-CM-05	Procurement to commissioning Commissioning	CC-PID-101 MEA-CM-PLN-001	**Materials TBC: passivation scope provisional** CC-PID-101 note 6 vs PC-03 The materials (note 6) are TBC, so the passivation scope (PC-03) is provisional.	Freeze the materials selection report.	**Open**
EQ-CM-06	Procurement to commissioning Commissioning	CC-PID-101 MEA-EE-LL-001 MEA-CM-PLN-001 also: SOP tab	**No spare pumps: changeover cannot be done** CC-PID-101 vs SOP register vs load list No spare pumps on the P&ID: the pump changeover in SOP-08 and commissioning step C-09 cannot be done as written. related: EQ-EE-02	Decide the sparing philosophy (EQ-EE-02).	**Open**
EQ-PX-01	P&ID (engineering view) Process	CC-PID-101 also: OTS model / HMI	**Measured flows not drawn on CC-PID-101** FI-303 (reflux) and FI-208 (make-up) are measured in the model but not drawn on CC-PID-101; PI-101, SI-101, TI-401 also not drawn (shown on faceplates).	Suggest FT bubbles at rev B.	**Open**
EQ-PX-02	P&ID (engineering view) Process	CC-PID-101 also: OTS model / HMI	**JI-101 on the sheet, not on the DCS overview** JI-101 is on the sheet but not on the DCS overview.	Add JI-101 (blower power) to the DCS overview or record it as a faceplate-only reading on the P&ID.	**Open**
EQ-PX-03	P&ID (engineering view) Process	CC-PID-101 also: OTS model / HMI	**E-104 / E-105 sheet duties without model counterpart** E-104 3.1 MW and E-105 5.8 MW duties on the sheet have no model counterpart to check against. related: EQ-EE-05	Check E-104 against the model lean cooler duty ($\dot{m}\cdot cp\cdot\Delta T$, HMB) and E-105 against the wash-water balance once the wash flow is set (EQ-EE-05).	**Open**
EQ-PX-04	P&ID (engineering view) Process	CC-PID-101	**PSVs and design pressures HOLD (note 5)** PSVs, vessel design pressures on HOLD (note 5); no spare pumps; no N2 connections (see EQ-CM-01/02, EQ-EE-02). related: EQ-CM-01, EQ-CM-02, EQ-EE-02	Close with the relief load basis (PSV-301, TSV-501…504, PSV-207 decision) and the EQL design pressures; show them at P&ID rev B.	**Open**
EQ-SA-01	Safety tabs (HAZOP, LOPA, SRS, Safety layers), collected here Safety	CC-PID-101 MEA-SA-HZP-001 MEA-PE-RLB-001 MEA-CM-PLN-001 also: OTS model / HMI	**PSVs on HOLD on the P&ID while other documents quote PSV-301** P&ID 5. Hold: "Psvs (t-102, v-101, e-102 shell, e-101), vessel design pressures and relief cases are not shown." The HMI text of PSHH-301 quotes PSV-301 at 350 kPa a and the relief load basis sizes it; the HAZOP never credits relief ("Relief valves HOLD (P&ID note 5) — not credited").	Issue the PSVs from MEA-PE-RLB-001 on CC-PID-101 rev B (PSV-301, TSV-501…504, the PSV-207 decision); then revalidate the HAZOP / LOPA rows that exclude relief.	**Open**

EQ	Raised by	Documents in conflict	Issue	Proposed resolution	Status
EQ-SA-02	Safety tabs (HAZOP, LOPA, SRS, Safety layers), collected here Safety	CC-PID-101 MEA-PE-RLB-001 MEA-ME-EQL-001 also: Safety layers tab	**P&ID note 5 puts a PSV on the "E-102 shell"** In a kettle reboiler the shell is the boiling process side, open to T-102 and covered by PSV-301; the steam is in the tubes. The Safety layers register calls the steam-side device PSV-207. Same finding as EQ-R01 and EQ-M05. related: EQ-R01, EQ-M05, EQ-R06	Revise note 5 as EQ-R01 proposes; record the E-102 steam-side decision (PSV-207 or MP re-rating, EQ-R06).	**Open**
EQ-SA-03	Safety tabs (HAZOP, LOPA, SRS, Safety layers), collected here Safety	MEA-SA-HZP-001 MEA-IN-SRS-001 MEA-IN-CE-001 MEA-IN-INL-001 CC-PID-101 also: LOPA tab	**T-102 sump high has no high-high trip** HAZOP N5-02 "High level in the T-102 sump" is ranked S3 × L4 = 12 (High) with alarms only; LOPA L6 needs a SIF (Add LSHH-208 on its own sensor: stop P-101 and shut a new XV-208 on the make-up water (SIL 1).). The SRS and the C&E carry it as LSHH-208 (SIF-6R, proposed, PFD 0.0301), but the instrument list and CC-PID-101 have no LSHH-208.	Close HAZOP R18 / LOPA LR8: add LSHH-208 on its own transmitter, XV-208 on the make-up water and the P-101 stop at P&ID rev B; add the rows to MEA-IN-INL-001.	**Open**
EQ-SA-04	Safety tabs (HAZOP, LOPA, SRS, Safety layers), collected here Safety	MEA-IN-SRS-001 MEA-IN-CE-001 MEA-PE-RLB-001 MEA-SA-HZP-001 also: LOPA tab	**SIF-2 PSHH-301: Gap in the SRS** Timing alone looks fine (PST 2.1 min, response 5.75 s), but with XV-207 shut the hot rich feed keeps flashing and the pressure still reaches 350 kPa a at 3.3 min (peak 400 kPa a): the SIF does not reach a safe state for a blocked outlet. Gap: SIF-2R (stop P-101) and PSV-301 are the LOPA fix. related: EQ-R04, EQ-M03	Implement SIF-2R (PSHH-301 also stops P-101, LOPA LR2) in the C&E and the SRS; keep PSV-301 sized for the blocked outlet (EQ-R04, EQ-M03).	**Open**
EQ-SA-05	Safety tabs (HAZOP, LOPA, SRS, Safety layers), collected here Safety	MEA-IN-SRS-001 MEA-IN-CE-001 MEA-IN-INL-001 MEA-EE-LL-001 also: OTS model / HMI	**Proposed SIF-7R (MCC under-voltage) has no instrument or electrical row** Trip I-2 directly on loss of power (MCC under-voltage relay → XV-207 shut, SIF-7R). Model: the pressure then stays at 175 kPa a instead of peaking near 350 kPa a. The SRS and the C&E list EUV-MCC (SIL 1 (SIF-7R, proposed)), but the instrument list has no row for it and the load list shows no under-voltage relay on MCC-101.	Add the under-voltage relay to the MCC-101 single-line / load list and an EUV-MCC row (DI to the SIS) to MEA-IN-INL-001.	**Open**

6 · Queries by document and discipline

Document	Title	Process	Safety	Instrumentation	Mechanical	Electrical	Commissioning	Total	Open
CC-PID-101	Piping and instrumentation diagram, absorber T-101 / strippe	10	3	2	4	4	5	28	28
MEA-PE-RLB-001	Relief load basis · PSV, TSV and vent register of the captur	12	3	2	1		1	19	17
MEA-ME-EQL-001	Equipment list and mechanical datasheets	4	1		8			13	11
MEA-IN-INL-001	Instrument list · I/O count · control-valve check	1	2	8				11	11
MEA-EE-LL-001	Electrical load list		1			6	1	8	8
MEA-PE-HMB-001	Heat and material balance, line sizing and pump hydraulics	6				1		7	7
MEA-CM-PLN-001	Procurement to commissioning plan		1				6	7	7
MEA-IN-SRS-001	Safety requirements specification of the safety instrumented	2	3		1			6	6
MEA-SA-HZP-001	HAZOP study report		3			1		4	4
MEA-IN-CE-001	Cause and effect matrix · SIS interlocks I-1 / I-2 / I-3 and		3					3	3
MEA-PFD-001	Process flow diagram	1						1	1
All queries		22	5	8	8	6	6	55	53

A query is counted once under every document it names, so the column totals exceed the number of queries. The P&ID is named most often: it is the hub of the tree, and the HOLDs of its note 5, the materials of note 6 and the equipment missing under note 7 reach every discipline.

7 · Checks

Check		Detail
MEA-PM-DOC-001: file link exists in the build list	✓	MEA-PM-DOC-001.pdf
MEA-PFD-001: file link exists in the build list	✓	MEA-PFD-001.json, MEA-PFD-001.pdf, MEA-PFD-001.png, MEA-PFD-001.pptx, MEA-PFD-001.svg
MEA-PE-HMB-001: file link exists in the build list	✓	MEA-PE-HMB-001.pdf
CC-PID-101: file link exists in the build list	✓	CC-PID-101.pdf, CC-PID-101.png, CC-PID-101.svg
MEA-SA-HZP-001: file link exists in the build list	✓	MEA-SA-HZP-001.pdf
MEA-IN-SRS-001: file link exists in the build list	—	no file issued (held in the SRS of SIF tab only)
MEA-IN-CE-001: file link exists in the build list	✓	MEA-IN-CE-001.pdf
MEA-ME-EQL-001: file link exists in the build list	✓	MEA-ME-EQL-001.pdf
MEA-PE-RLB-001: file link exists in the build list	✓	MEA-PE-RLB-001.pdf
MEA-IN-INL-001: file link exists in the build list	✓	MEA-IN-INL-001.pdf
MEA-EE-LL-001: file link exists in the build list	✓	MEA-EE-LL-001.pdf
MEA-CM-PLN-001: file link exists in the build list	✓	MEA-CM-PLN-001.pdf
Every file in the build list belongs to a register document	✓	17 files
Every EQ references at least one register document	✓	55/55; 10 only through the raising document
Every document an EQ cites is in the register	✓	all
EQ numbers are unique	✓	55 numbers
Every cross-referenced EQ exists	✓	16 cross-references
Every EQ has a proposed resolution	✓	8 proposed by this register
ecomm: NOTES.md queries = the tab's data file	✓	6 queries
eelec: NOTES.md queries = the tab's data file	✓	6 queries
einst: NOTES.md queries = the tab's data file	✓	8 queries
emech: NOTES.md queries = the tab's data file	✓	8 queries
eproc: NOTES.md queries = the tab's data file	✓	6 queries
erel: NOTES.md queries = the tab's data file	✓	12 queries
pidx/NOTES.md: queries collected	✓	EQ-PX-01, EQ-PX-02, EQ-PX-03, EQ-PX-04
The deliverable tree has no loop (input links)	✓	PFD → Safety layers → HMB → P&ID → Equipment list → HAZOP report → Load list → LOPA → Relief load basis → SRS of SIF → Instrument list → C&E matrix → Commissioning plan
Every register document is on the tree	✓	11 documents
No document is first issued before a document that feeds it	✓	all links in phase order

27 ✓, 0 ✗. The tab repeats these checks in the page and adds the live comparison of each tab's own data file with this register when that tab is in the page. Data made 2026-10-28 06:08.